PaperSonar

Fact and dimension identification

Design baseline 1.0 / 25 October 2026

PaperSonar models three different processes: publishing a scholarly work, observing its current citation and access state, and recording a directed reference relationship. Separating these processes avoids mixing publication output with repeated observations or citation edges.

Business processes and fact types

Process	Fact table	Type and grain
Publication output	fact_publication	Transaction fact. One admitted canonical work.
Citation and access observation	fact_work_snapshot	Periodic snapshot. One work on one successful UTC observation day.
Reference relationship	fact_citation	Factless fact. One distinct directed citing and cited work pair.

The publication event is indexed using its source-reported publication date. A later metadata correction updates that work's current event attributes. It does not create another publication. Observation snapshots are recorded when the source is fetched, while the citation relationship records when PaperSonar first saw a pair.

Measures and valid aggregation

Table	Measure	Aggregation rule
Publication	publication_count	Always 1. Sum over the publication grain. Deduplicate after any multivalued bridge join.
Publication	author_count and reference_count	Counts from source metadata. Sum only for aggregate involvement or reference volume. Average per work only after work-level deduplication.
Snapshot	cited_by_count	Sum across distinct works at one selected snapshot date. Do not sum over observation dates.
Snapshot	observed_count	Always 1. Counts observations, not new publications.
Citation	No stored measure	Count distinct directed pairs. A COUNT result is derived, so the relationship fact remains factless.

Dimensions and hierarchies

Dimension	Natural identity	Main attributes and hierarchy
dim_work	Canonical OpenAlex work ID	DOI, title, publication year, language, precision, scope and stub flags.
dim_date	Calendar date	Day to month to quarter to year. Role-played for publication, observation and first-seen dates.
dim_topic	OpenAlex topic ID	Topic to subfield to field to domain. Higher levels are flattened attributes.
dim_venue	OpenAlex source ID	Name, ISSN-L and source type. No unsupported venue quality rank.
dim_work_type	Source type code	Publication type code and readable label.
dim_access_status	Source access status code	Status and nullable open-access indicator.
dim_author	OpenAlex author ID	Display name and supplied ORCID. Names are not identity keys.
dim_institution	OpenAlex institution ID	Name, ROR ID, type and country. Institution to country roll-up.

All dimensions use integer surrogate keys. Date uses YYYYMMDD as its surrogate convention, with 0 for unknown. OpenAlex IDs are retained as unique alternate keys. DOI is a nullable unique alternate identifier after conflict resolution. A title or author name never becomes a natural key.

Slowly changing attributes

The first release uses Type 1 updates for current descriptive attributes and work-topic, work-author and work-institution memberships. Raw inputs and dated report exports preserve the evidence for old comparisons. The model does not claim that joining an old snapshot to current topic assignments reproduces a past classification state. Rebuild from archived inputs when that state is required. A versioned membership model is a future extension.

Bridges and conformed use

Bridge	Relationship and policy
bridge_work_topic	One work to many supplied topics. Retain source score and primary flag. Equal allocation weights sum to 1 across all distinct supplied topics for the work.
bridge_work_author	One work to many distinct supplied authors. Retain source author position. Institution attribution is not joined through this bridge in reporting.
bridge_work_institution	One work to many distinct institutions deduplicated across authors. Equal allocation weights sum to 1 across all distinct institutions for the work.

Weights are calculated over the complete supplied set before a report filter is applied. Do not renormalize the remaining weights after filtering institutions or topics. An absent membership set remains absent and is counted by a separate missingness measure. A source assignment score is not an allocation weight.

Conformed dimensions and report paths

The work and date dimensions are shared across publication and snapshot analysis. dim_work is role-played twice for citation analysis. For Q11, join each citation endpoint to its own publication fact alias to obtain the endpoint's primary topic. Citation targets that exist only as stubs are excluded from topic-flow analysis and counted separately as unresolved or out-of-scope targets.

There are no direct foreign keys between fact tables. Combine facts only after selecting their intended grain and observation date. For publication output by primary topic, read fact_publication directly. For secondary-topic exploration, use the topic bridge and a distinct work count or fractional weights. Never multiply author, institution and topic bridges in one unrestricted join.

Run lineage

etl_run is an operational lineage table rather than a business dimension. It records corpus, mapping and metric versions, manifest path, timestamps, status and source-record counts. It is shown separately from the analytical constellation. Publication and observation rows retain a run FK, and citation pairs retain their last complete observation run.